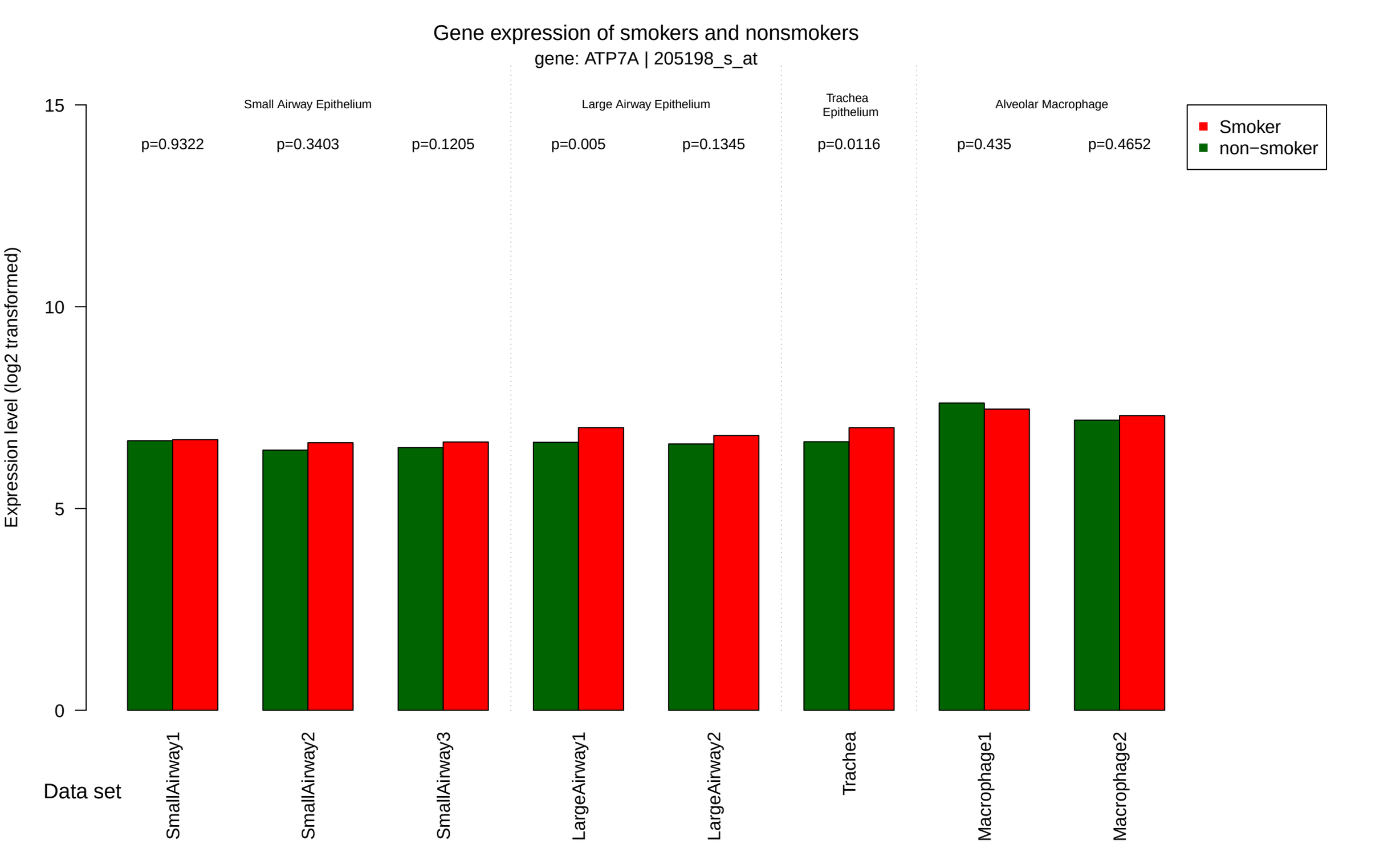
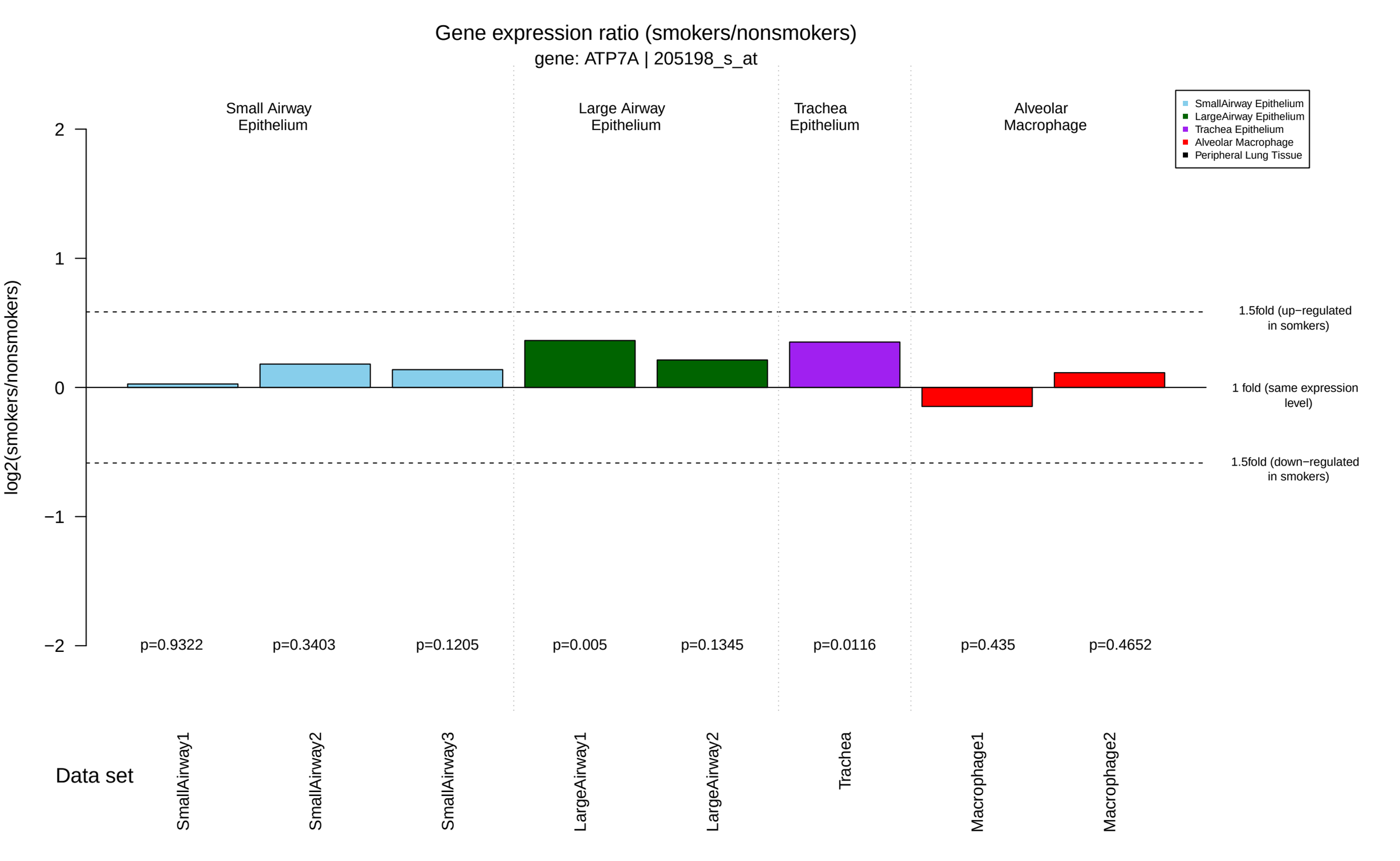
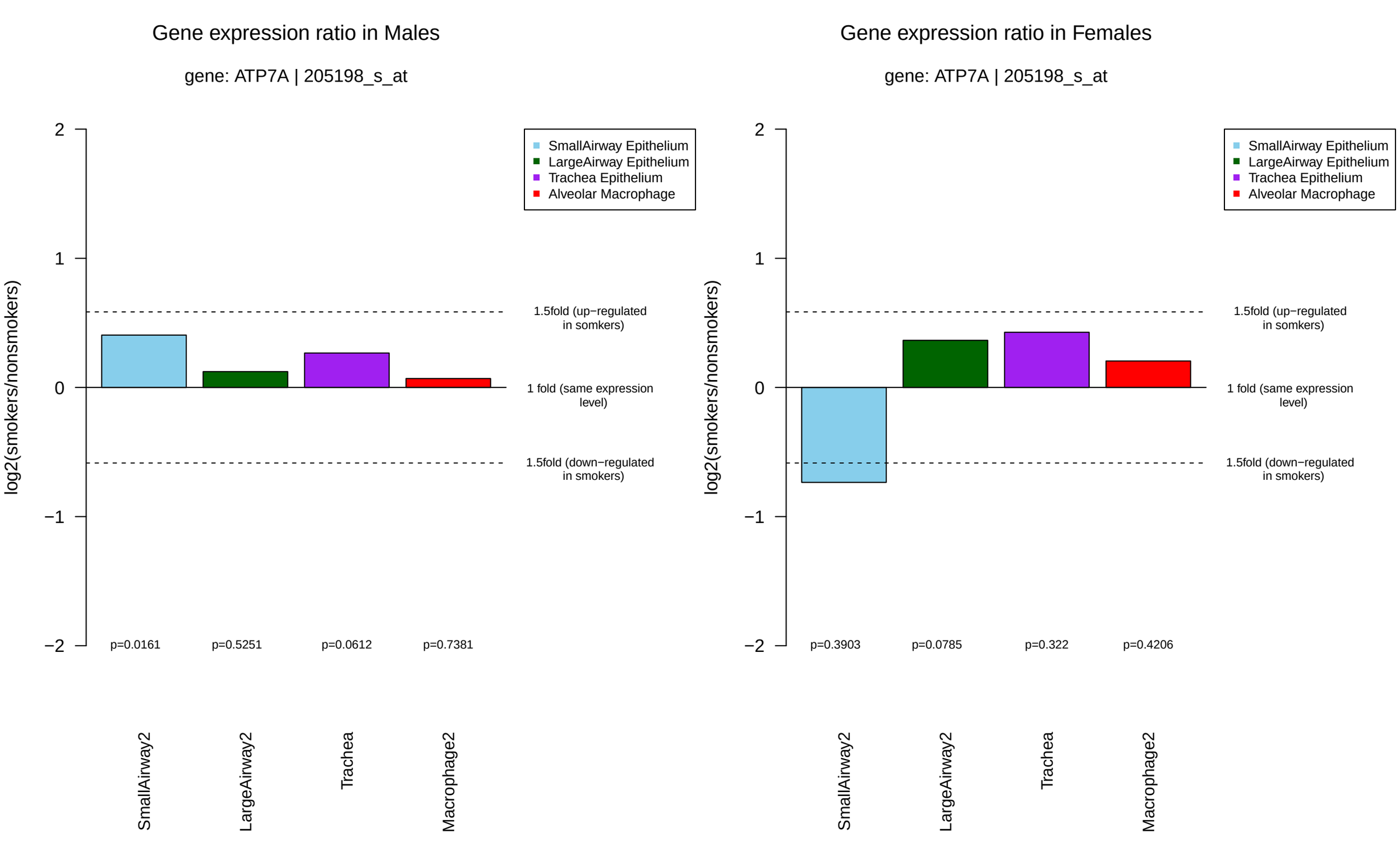
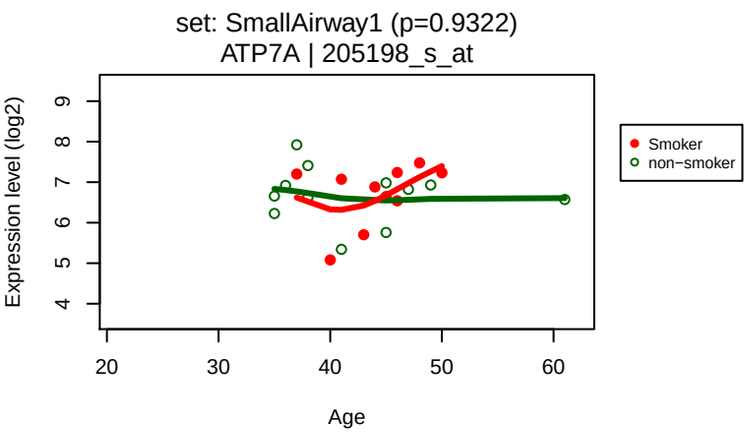
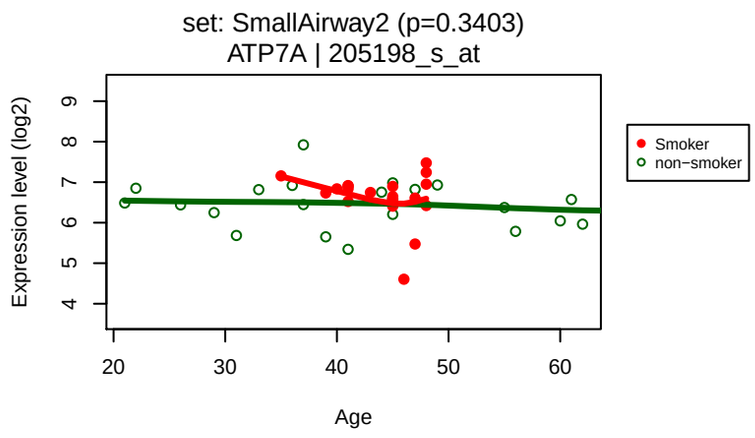
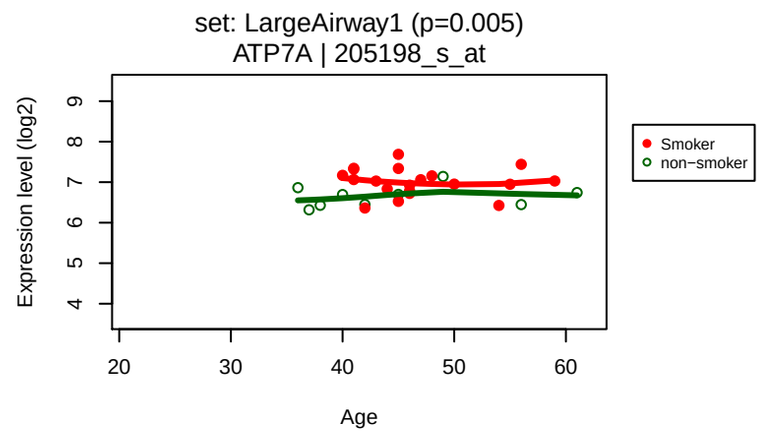

Gene expression of smokers and nonsmokers
gene: ATP7A | 205198_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9322
p=0.3403
p=0.1205
p=0.005
p=0.1345
p=0.0116
p=0.435
p=0.4652
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

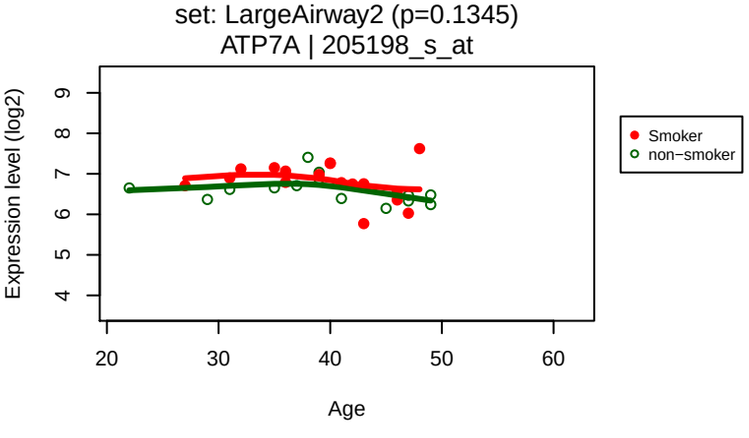

Gene expression ratio (smokers/nonsmokers)
gene: ATP7A | 205198_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9322
p=0.3403
p=0.1205
p=0.005
p=0.1345
p=0.0116
p=0.435
p=0.4652
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

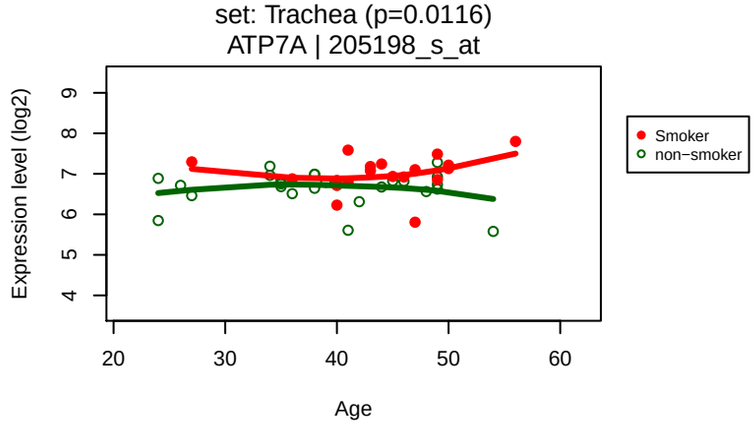

Gene expression ratio in Males
gene: ATP7A | 205198_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0161
p=0.5251
p=0.0612
p=0.7381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP7A | 205198_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3903
p=0.0785
p=0.322
p=0.4206
SmallAirway2
LargeAirway2
Trachea
Macrophage2

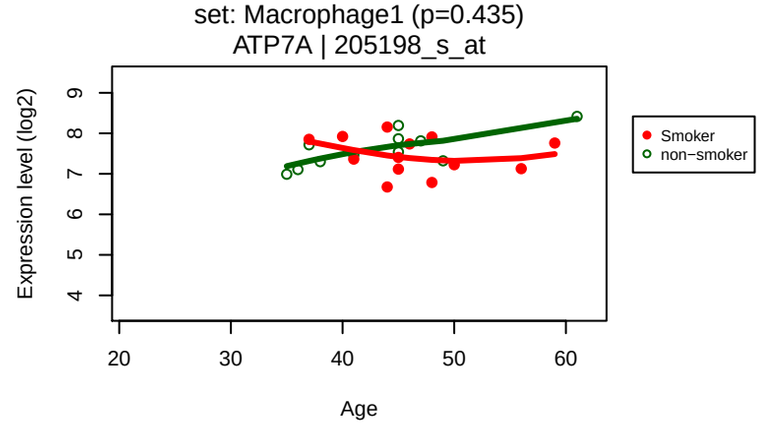

set: SmallAirway1 (p=0.9322)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker

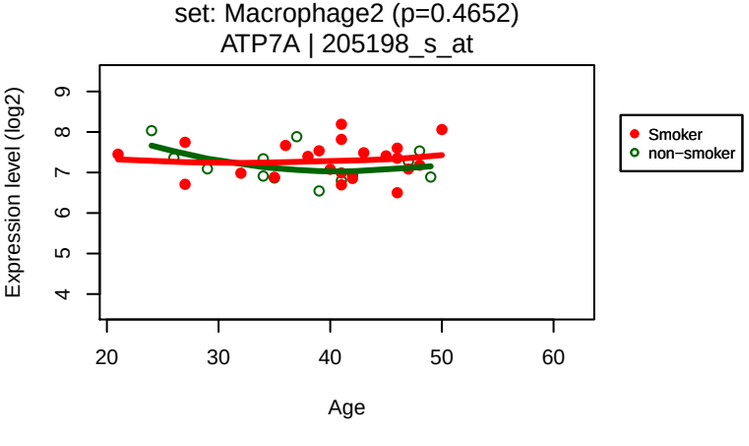

set: SmallAirway2 (p=0.3403)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker

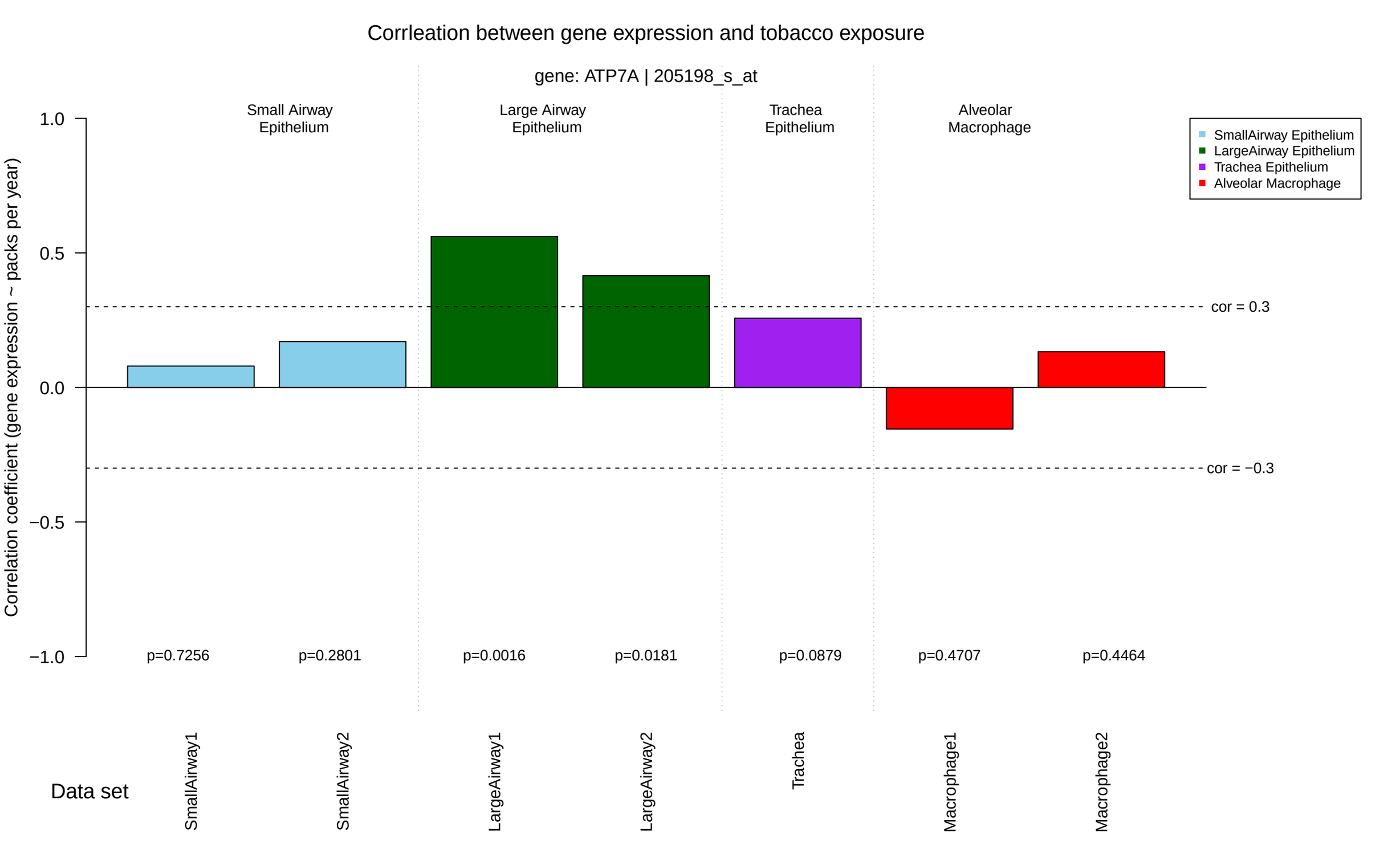

set: LargeAirway1 (p=0.005)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1345)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0116)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.435)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4652)
ATP7A | 205198_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP7A | 205198_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7256
p=0.2801
p=0.0016
p=0.0181
p=0.0879
p=0.4707
p=0.4464
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2